

# GATI VARSHNEY

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## Education

**JSS Academy of Technical Education**, Noida, India

*Bachelor of Technology in Computer Science and Engineering (Data Science)*

2023 – 2027

**GPA: 8.7/10**

## Experience

**Software Engineer Intern**, Quin Que Inc. (Japan, Remote)

Sep 2026

- Shipped 6 admin screens for Rapilogi, a multi-tenant warehouse platform, across the OpenAPI contract, PostgreSQL migrations, Go handlers and React, with MFA login and Row-Level Security for tenant isolation.
- Caught a tenant-provisioning defect in the design documents before writing code, and generalized the fix into a reusable tenant-preset field instead of a one-off patch.
- Fixed SSE reconnection and React Native crashes outside assigned scope; 4 PRs merged through code review.

**Google Summer of Code 2026 Contributor**, The Linux Foundation

May 2026 – Aug 2026

*AI-Driven Printer Compatibility & Recommendation Portal* [\[Final Report\]](#)

- Built a recommendation engine over 6,657 printers that loads 6 KB per page instead of 24 MB by ranking them on 415 features with IDF-weighted cosine similarity and sharding output per printer.
- Diagnosed that 86.8% of similarity scores saturated at 1.0, making rankings meaningless; cut it to 0% with evidence damping and capability-conflict penalties, guarded by a harness that fails on any metric drift.
- Engineered a natural-language printer assistant with zero page-load cost, verified by 384 tests incl. a 187-utterance corpus, by running deterministically in-browser and flagging data gaps explicitly.

**Open Source Contributor**, OpenPrinting [\[Merged PRs\]](#)

Nov 2025 – Aug 2026

- Delivered production site search across 200+ posts with no search server by compiling posts into a static JSON index via a build-time AST pipeline (unified, remark-parse), queried in-browser by MiniSearch with fuzzy matching.
- Migrated 200+ Jekyll posts and 118 media files to Next.js with legacy-URL redirects, and shipped the author system and homepage; 14 PRs merged through maintainer review.
- Fixed production 404s across 7.5K+ routes, including RSS links consumed by LWN, by tracing them to a deploy-time config overriding the repo's, and added a CI check blocking deploys with broken internal links.

## Projects

**Contributable** | Next.js, TypeScript, GraphQL, GitHub Actions [\[Live\]](#) [\[GitHub\]](#)

2026

- Launched a live platform ranking 1,500+ repositories from Google Summer of Code organizations by newcomer-friendliness, measured on first-reply time, outside-contributor merge rate and open starter issues.
- Served fresh data with no database and no GitHub call on any page view by scheduling a GitHub Actions sweep of the GraphQL API that fetches only changes and commits snapshots to a data branch.
- Made every published figure defensible by modeling reply and merge times with Kaplan-Meier survival estimates, filtering bots, withholding any figure under 5 samples and linking each number to its source.

**cidx** | Python, tree-sitter, SQLite, MCP [\[Live\]](#) [\[GitHub\]](#)

2026

- Published a zero-config code index for AI coding agents to PyPI, validated on Django (3,000+ files), by parsing Python and TypeScript/JavaScript with tree-sitter into a SQLite symbol index served over MCP.
- Guaranteed incremental updates match a cold rebuild exactly by enforcing content-hash invalidation and per-file SQLite transactions, verified by Hypothesis property tests and crash injection in CI on 3 operating systems.

## Achievements

- Winter of Code 5.0 (IIIT Kalyani's open-source program): Ranked Top 20 of 2,800+ participants
- OpenPrinting Hall of Fame: Recognized by maintainers for contributions to the project website [\[Listing\]](#)
- LeetCode Knight: Contest rating 1925, top 4% of contest participants globally [\[Profile\]](#)

## Technical Skills

**Languages:** C++, Python, TypeScript, JavaScript, SQL

**Frontend:** React.js, Next.js, Tailwind CSS, HTML5, CSS3

**Backend:** Node.js, Express.js, REST APIs, GraphQL, OpenAPI, PostgreSQL, MongoDB, SQLite, JWT & NextAuth

**AI Integration:** Google Gemini API, Model Context Protocol (MCP), schema-constrained structured outputs (Zod)

**Developer Tools:** Git, GitHub Actions (CI/CD), Linux, tree-sitter, pytest, Hypothesis, Puppeteer

**Core Subjects:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks